

Skander Moalla

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EDUCATION

EPFL

Lausanne, Switzerland

PhD in Computer and Communication Sciences Sep 2022 – Present (graduation expected in Fall 2026)
Learning from Experience: Plasticity, Reinforcement, and Exploration. Advised by Prof. Caglar Gulcehre.
Deep reinforcement learning, LLM post-training, diversity and test-time scaling, agentic formal math proving.

University of Oxford

Oxford, UK

MSc in Advanced Computer Science Oct 2021 – Sep 2022
MSc thesis in multi-agent reinforcement learning with Prof. Shimon Whiteson's group (WhiRL).

École Polytechnique

Paris, France

BSc in Mathematics and Computer Science (Double Major) Sep 2017 – Aug 2020
Graduated Summa Cum Laude — CGPA: 4.09/4.0 — Awarded an excellence scholarship.
Exchange programs: University of Toronto (Fall 2019) & Stanford University (Summer 2019).

EXPERIENCE

Meta FAIR

Paris, France

Research Scientist Intern Mar 2026 – Aug 2026
○ Hosted by Julia Kempe and Rémi Munos in the Foundations of Reasoning / Math Superintelligence Team.
○ Worked on agentic in-context adaptation, self-improvement, and asymmetric self-play in formal math (Lean).

Google DeepMind

Paris, France

PhD Student Researcher Sep 2025 – Dec 2025
○ Hosted by Alexandre Ramé in the Gemma post-training team (Olivier Bachem, Lucas Dixon).
○ Worked on leveraging diversity to improve test-time scaling (pass@k) and sampling in reinforcement learning.
○ Contributed to the Gemma open-source JAX codebase (Gemma 3 KV cache sharing and sampler).

PUBLICATIONS

Project Apertus, [...], **Skander Moalla***, [...], Antoine Bosselut, Martin Jaggi, and Imanol Schlag. Apertus: Democratizing Open and Compliant LLMs for Global Language Environments. *ACL*, 2026.

Simon Matrenok*, **Skander Moalla***, and Caglar Gulcehre. Quantile Reward Policy Optimization: Alignment with Pointwise Regression and Exact Partition Functions. *NeurIPS*, 2025.

Skander Moalla, Andrea Miele, Razvan Pascanu, and Caglar Gulcehre. No Representation, No Trust: Connecting Representation, Collapse, and Trust Issues in PPO. *NeurIPS*, 2024.

Xiuying Wei, **Skander Moalla**, Razvan Pascanu, and Caglar Gulcehre. Building on Efficient Foundations: Effectively Training LLMs with Structured Feedforward Layers. *NeurIPS*, 2024.

Skander Moalla*, Manuel Madeira*, Lorenzo Riccio*, and Joonhyung Lee*. [Re] Reproducibility Study of Behavior Transformers. *ReScience C*, 9(2), 2023. **Outstanding Paper Awards Honorable Mentions.**

Benjamin Ellis, Jonathan Cook, **Skander Moalla**, Mikayel Samvelyan, Mingfei Sun, Anuj Mahajan, Jakob Foerster, and Shimon Whiteson. SMACv2: An Improved Benchmark for Cooperative Multi-Agent Reinforcement Learning. *NeurIPS*, 2023.

EXPERIENCE (CONTINUED)

Quincus

Applied Research Intern

Singapore (Remote)

Mar 2021 – Aug 2021

- Deployed a solution leveraging deep reinforcement learning (RL) to optimize middle-mile logistics.
- Built simulators to model multi-mile parcel shipping and trained deep RL prototypes (PPO, DQN).
- Stack: Python (OpenAI Gym & Baselines, LightGBM, pytest), Docker, NSCC High-Performance Computing.

Amazon

Software Development Intern

Luxembourg

Sep 2020 – Feb 2021

- Completed full-stack and DevOps sprint tasks within a “two-pizza team” owning multiple services end-to-end.
- Designed and built a prototype to detect similar and duplicate business metrics supporting Amazon’s transportation network worldwide to ease metric discovery and reduce computation costs leveraging deep learning and NLP techniques with AWS SageMaker.
- Stack: AWS, Python, Java (Dagger, AutoValue, JUnit, Mockito), TypeScript (ImmutableJS, Mocha, Chai).

EPFL – Laboratory for Topology and Neuroscience & Blue Brain Project

Geneva, Switzerland

Research Intern

Jan 2020 – Apr 2020

- BSc thesis supervised by Prof. Kathryn Hess Bellwald ([Link to thesis](#)).
- Built a framework for neuron simulation and leveraged tools from information theory and algebraic topology (simplicial homology) to formulate new conclusions about the dynamics of neurons forming directed cliques.
- Stack: Python 3 (DIT, JupyterLab, NumPy, SciPy, Brian2), C++.

Ooredoo

Full-Stack Developer Intern

Tunis, Tunisia

Jul 2018 – Aug 2018

- Implemented a PoC for a custom notification system in a hybrid mobile app featuring profiling functionalities and allowing Ooredoo to adjust the activity of their notification server. Stack: Ionic 3, Node.js, socket.io.

SOFTWARE

Core stack and responsibilities.....

Co-lead of the Swiss AI Apertus v1 post-training

July 2024 – Present

- Developed infrastructure and offline post-training code to train the Swiss open models Apertus 8B and 70B on the CSCS Alps cluster with 1000+ GH200 nodes.
- Supervised MS students on chat templates, evaluations, supervised fine-tuning, multilinguality, and data.

Point of Contact for infrastructure & compute platforms in the lab

Sep 2023 – Present

- Support the deployment of container-based research experiments on multiple clusters at the scale of hundreds of nodes, including Slurm HPC clusters and Run:AI (Kubernetes-based) clusters. ([Documentation](#))
- Stack: Docker, Kubernetes, Run:AI, Slurm, NVIDIA Pyxis, NVIDIA Enroot, Apptainer/Singularity.

Open-source contributions and selected projects.....

Open-source contributions

- @PyTorch/RL, @DLR-RM/stable-baselines3, @Farama-Foundation/Gymnasium, @HuggingFace/TRL.

Python Machine Learning Research Project Template

Sep 2022 – Present

- Maintain a template for reproducible Python machine learning research projects deployable on various platforms including Slurm (with containers through NVIDIA Pyxis/Enroot or Apptainer) and Kubernetes clusters.

TEACHING

Université Dauphine Tunis (Université Paris Dauphine-PSL)

Tunis, Tunisia

Visiting Lecturer – Reinforcement Learning

Jan 2026

- Delivered a 24-hour course for the Master’s in Big Data & Artificial Intelligence (BDIA).
- Covered reinforcement learning foundations and modern LLM post-training methods such as RLHF.

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| EPFL | Lausanne, Switzerland |
| <i>LauzHack Deep Learning Bootcamp Instructor (Deep Reinforcement Learning)</i> | <i>Jul 2024</i> |
| ○ Delivered a 2-hour introductory lecture on deep reinforcement learning (Bellman equations, DQN, PPO, etc.). | |
| EPFL | Lausanne, Switzerland |
| <i>Teaching Assistant & Supervisor</i> | <i>Sep 2022 – Present</i> |
| ○ Supervised 9 Master's projects and theses. | |
| ○ Probability and Statistics (MATH-232 Spring 2023), Machine Learning (CS-433 Fall 2023 – exam preparation), Artificial Neural Networks / Reinforcement Learning (CS-456 Spring 2024 – project preparation). | |
| ATSM – Tunisian Association of Mathematics | Tunis, Tunisia |
| <i>Olympiad Coach</i> | <i>Jul 2018 – Present</i> |
| ○ Designed material for Olympiad problem-solving and trained national team members for the Mediterranean Mathematical Olympiad (MYMC 2018) and the International Mathematical Olympiad (IMO 2020). | |

SOFTWARE - CLASS PROJECTS

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| SAM - Study Abroad Matching | <i>Sep 2019 – Dec 2019</i> |
| ○ Designed a web app around an algorithm to match students applying to study-abroad programs and universities based on a human-in-the-loop solution to the Gale–Shapley college admissions problem. | |
| ○ Stack: Python, Django, React, Google Cloud Platform (GCP), Google App Engine, Google Cloud SQL. | |
| Computer Facial Discrimination - Machine Learning Project | <i>May 2019 – Jun 2019</i> |
| ○ Studied the effect of different pre-trained feature extractors (Autoencoders, CNN classifiers) and clustering methods on the discrimination of human faces. | |
| ○ Built a customizable autoencoder model to demonstrate that training data (as an analogy for background and culture) can be the root of multiple kinds of discrimination. | |
| ○ Stack: Python, TensorFlow, Keras, scikit-learn, Google Colab, Kaggle. | |
| TOP5 Basketball Team Manager - C++ Project | <i>Nov 2018 – Jan 2019</i> |
| ○ Managed a team of 9 students / Led the Game Engine team / Maintained the Git repository. | |
| ○ Designed and implemented the game engine and its interaction with the remaining game entities. | |
| ○ Stack: C++, Qt. | |
| Meet Halfway - Web Project | <i>May 2018 – Jun 2018</i> |
| ○ Built a user-friendly website that allows a group of friends to find the best destination to travel to and meet for holidays according to their respective locations and cost of travel. | |
| ○ Stack: HTML, CSS, JS. | |

EDUCATION - EXCHANGE PROGRAMS

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| University of Toronto | Toronto, Canada |
| <i>Exchange – Fall 2019 — Software Engineering, Databases, Compilers & Interpreters</i> | <i>Sep 2019 – Dec 2019</i> |
| Stanford University | Stanford, USA |
| <i>Exchange – Summer 2019 — GPA: 4.15/4.0 — AI, Entrepreneurship & Innovation</i> | <i>Jun 2019 – Aug 2019</i> |

LANGUAGES

English: Bilingual French: Native Arabic: Native German: Elementary